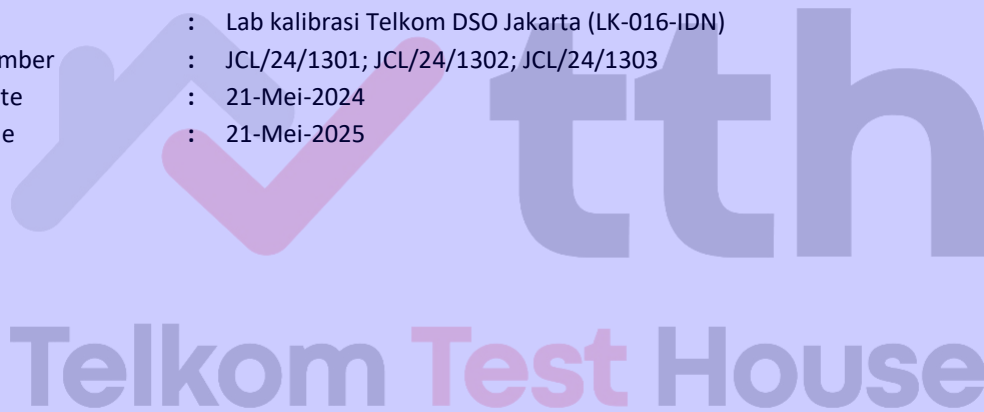


EXTERNAL CALIBRATION CERTIFICATE

Applicant's Data : PT Telkom Indonesia (Persero), Tbk
Jl. Japati No. 1 Sadangserang, Coblong, Kota Bandung, Jawa Barat .

Equipment Data
Equipment name : OPTICAL LIGHT SOURCE
Brand : EXFO
Type/Model : 2150
Serial number : 1656285
Asset number : (217/CAL)

Calibration Information
Calibration By : Lab kalibrasi Telkom DSO Jakarta (LK-016-IDN)
Certificate Number : JCL/24/1301; JCL/24/1302; JCL/24/1303
Calibration Date : 21-Mei-2024
Calibration Due : 21-Mei-2025



Agreed by: **Bandung, 13 Juni 2024**

Reviewed by:

I GEDE ASTAWA
OSM Infrastructure Research & Assurance

ADI PERMADI
Manager Lab Infrastructure QA



No. Dokumen : Form TCL-QP-19-01

No. / Tgl. Terbit : 01 / 20 Juli 2017

No. / Tgl. Revisi : 02 / 04 Januari 2021



CERTIFICATE OF CALIBRATION

P T TELKOM INDONESIA Tbk
KAN ACCREDITED CALIBRATION LABORATORY (LK-016-IDN)

Jakarta : Jl. Percetakan Negara No.17-19 Jakarta Pusat 10570 Phone 021-21480341
Medan : Jl. Gaharu No. 1 Medan 20235 Phone 061-4531311 Fax. 061-4147434
Surabaya : Jl. Gayungan PTT No. 17-19 Surabaya 60235 Phone 031-8283678 Fax. 031-8283388
Makassar : Jl. AP Pettarani No. 4 Makassar 90221 Phone 0411-861153

Certificate Number

JCL/24/1301/DY

Approved Signatory

M. Ilyas

Date of Issue : 27-May-2024
Submitted by : PT. TELKOM INDONESIA Tbk.
DIGITAL CONNECTIVITY SERVICE
Attention : Manager Infrastructure Quality Assurance
Description : Optical Light Source
Manufacturer/Model : EXFO / FTBx-2150-0236B-3EA
Serial Number : 1656285 / Wavelength λ 1310 nm (CH 1) / (217/CAL)
Environment : $22.6^{\circ}\text{C} \pm 0.2^{\circ}\text{C}$ 56 % $\pm$ 0.3 % RH
Calibration Location : In-House
Date Calibrated : 21-May-2024
Recommended Due Date : 20-May-2025
Date Received : 21-May-2024

PT. Telkom Indonesia, Tbk. - Telkom Certifies that this instrument has been calibrated under the environmental condition stated above using telkom standard (s) which is (are) traceable to the international system of Units (SI) through SCL Hong Kong.

Method of Calibration :

The calibration was carried out based on the established Standard Telkom Calibration Procedure, TCL-WI-0012 and by comparison against the following standard(s) and instrument(s) which is (are) calibrated on schedule to maintain traceability and required accuracy :

Description	S/N	Cal Date	Due Date
1. Optical Fiber Power Meter	1030788-CH1	24-Dec-2021	23-Dec-2024

CERTIFICATE OF CALIBRATION

Checked By : *NU*

Certificate Number : JCL/24/1301/DY

1. NOMINAL WAVELENGTH λ 1310.0 nm

1.2. STABILITY TEST RESULT

	Result	
MIN Power	4.7844	dBm
MAX Power	4.7931	dBm
Delta Power	0.0087	dB

Detailed Data

Time (s)	Value (dBm)
0	4.7844
30	4.7858
60	4.7875
90	4.7894
120	4.7919
150	4.7931
180	4.7912
210	4.7897
240	4.7895
270	4.7916
300	4.7905
330	4.7923

ESTIMATED UNCERTAINTY

Test	Estimated uncertainty
1. Level Test 1310.0 nm	± 0.25 dB

No. Dokumen : Form TCL-QP-19-01

No. / Tgl. Terbit : 01 / 20 Juli 2017

No. / Tgl. Revisi : 02 / 04 Januari 2021

CERTIFICATE OF CALIBRATION

P T TELKOM INDONESIA Tbk
KAN ACCREDITED CALIBRATION LABORATORY (LK-016-IDN)

Jakarta : Jl. Percetakan Negara No.17-19 Jakarta Pusat 10570 Phone 021-21480341
Medan : Jl. Gaharu No. 1 Medan 20235 Phone 061-4531311 Fax. 061-4147434
Surabaya : Jl. Gayungan PTT No. 17-19 Surabaya 60235 Phone 031-8283678 Fax. 031-8283388
Makassar : Jl. AP Pettarani No. 4 Makassar 90221 Phone 0411-861153

Certificate Number

JCL/24/1302/DY

Approved Signatory


M. Ilyas

Date of Issue : 27-May-2024
Submitted by : PT. TELKOM INDONESIA Tbk.
DIGITAL CONNECTIVITY SERVICE
Attention : Manager Infrastructure Quality Assurance
Description : Optical Light Source
Manufacturer/Model : EXFO / FTBx-2150-0236B-3EA
Serial Number : 1656285 / Wavelength λ 1490 nm (CH 2) / (217/CAL)
Environment : 22.6 °C $\pm$ 0.2 °C 56 % $\pm$ 0.3 % RH
Calibration Location : In-House
Date Calibrated : 21-May-2024
Recommended Due Date : 20-May-2025
Date Received : 21-May-2024

PT. Telkom Indonesia, Tbk. - Telkom Certifies that this instrument has been calibrated under the environmental condition stated above using telkom standard (s) which is (are) traceable to the international system of Units (SI) through SCL Hong Kong.

Method of Calibration :

The calibration was carried out based on the established Standard Telkom Calibration Procedure, TCL-WI-0012 and by comparison against the following standard(s) and instrument(s) which is (are) calibrated on schedule to maintain traceability and required accuracy :

Description	S/N	Cal Date	Due Date
1. Optical Fiber Power Meter	1030788-CH1	24-Dec-2021	23-Dec-2024

CERTIFICATE OF CALIBRATION

Checked By : *[Signature]*

Certificate Number : JCL/24/1302/DY

1. NOMINAL WAVELENGTH λ 1490.0 nm (diukur dengan menggunakan λ 1550.0 nm)

1.2. STABILITY TEST RESULT

	Result	
MIN Power	4.9446	dBm
MAX Power	4.9509	dBm
Delta Power	0.0063	dB

Detailed Data

Time (s)	Value (dBm)
0	4.9507
30	4.9509
60	4.9508
90	4.9505
120	4.9485
150	4.9470
180	4.9454
210	4.9446
240	4.9450
270	4.9448
300	4.9448
330	4.9448

ESTIMATED UNCERTAINTY

Test	Estimated uncertainty
1. Level Test 1310.0 nm	± 0.25 dB

No. Dokumen : Form TCL-QP-19-01

No. / Tgl. Terbit : 01 / 20 Juli 2017

No. / Tgl. Revisi : 02 / 04 Januari 2021



CERTIFICATE OF CALIBRATION

P T TELKOM INDONESIA Tbk
KAN ACCREDITED CALIBRATION LABORATORY (LK-016-IDN)

Certificate Number

JCL/24/1303/DY

Approved Signatory

Jakarta : Jl. Percetakan Negara No.17-19 Jakarta Pusat 10570 Phone 021-21480341
Medan : Jl. Gaharu No. 1 Medan 20235 Phone 061-4531311 Fax. 061-4147434
Surabaya : Jl. Gayungan PTT No. 17-19 Surabaya 60235 Phone 031-8283678 Fax. 031-8283388
Makassar : Jl. AP Pettarani No. 4 Makassar 90221 Phone 0411-861153

M. Ilyas

Date of Issue : 27-May-2024
Submitted by : PT. TELKOM INDONESIA Tbk.
DIGITAL CONNECTIVITY SERVICE
Attention : Manager Infrastructure Quality Assurance
Description : Optical Light Source
Manufacturer/Model : EXFO / FTBx-2150-0236B-3EA
Serial Number : 1656285 / Wavelength λ 1550 nm (CH 3) / (217/CAL)
Environment : 22.6 °C $\pm$ 0.2 °C 56 % $\pm$ 0.3 % RH
Calibration Location : In-House
Date Calibrated : 21-May-2024
Recommended Due Date : 20-May-2025
Date Received : 21-May-2024

PT. Telkom Indonesia, Tbk. - Telkom Certifies that this instrument has been calibrated under the environmental condition stated above using telkom standard (s) which is (are) traceable to the international system of Units (SI) through SCL Hong Kong.

Method of Calibration :

The calibration was carried out based on the established Standard Telkom Calibration Procedure, TCL-WI-O012 and by comparison against the following standard(s) and instrument(s) which is (are) calibrated on schedule to maintain traceability and required accuracy :

Description	S/N	Cal Date	Due Date
1. Optical Fiber Power Meter	1030788-CH1	24-Dec-2021	23-Dec-2024

CERTIFICATE OF CALIBRATION

Checked By : 

Certificate Number : JCL/24/1303/DY

1. NOMINAL WAVELENGTH λ 1550.0 nm

1.2. STABILITY TEST RESULT

	Result	
MIN Power	5.1250	dBm
MAX Power	5.1298	dBm
Delta Power	0.0048	dB

Detailed Data

Time (s)	Value (dBm)
0	5.1295
30	5.1288
60	5.1284
90	5.1250
120	5.1252
150	5.1260
180	5.1268
210	5.1280
240	5.1288
270	5.1298
300	5.1294
330	5.1292

ESTIMATED UNCERTAINTY

Test	Estimated uncertainty
1. Level Test 1310.0 nm	± 0.25 dB